

sdstudio

A Companion Package for Designing
and Managing surveydown Surveys

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About Me



- **Pingfan Hu**, PhD Candidate at George Washington University, advised by Dr. John Helveston.
- Research interests:
 1. **Data Science** in sustainable transportation and human behavior
 2. **Open-Source Software** in  tools for survey research
 3. **Agentic AI Engineering** around Claude Code
- For more information, visit pingfanhu.com

jhelvy.shinyapps.io

 **Welcome to the Ultimate Superhero Survey!** 

Thank you for participating in our research on superhero preferences and powers! This survey will help us understand what makes superheroes appealing to different people.

This survey should take about 3-5 minutes to complete.

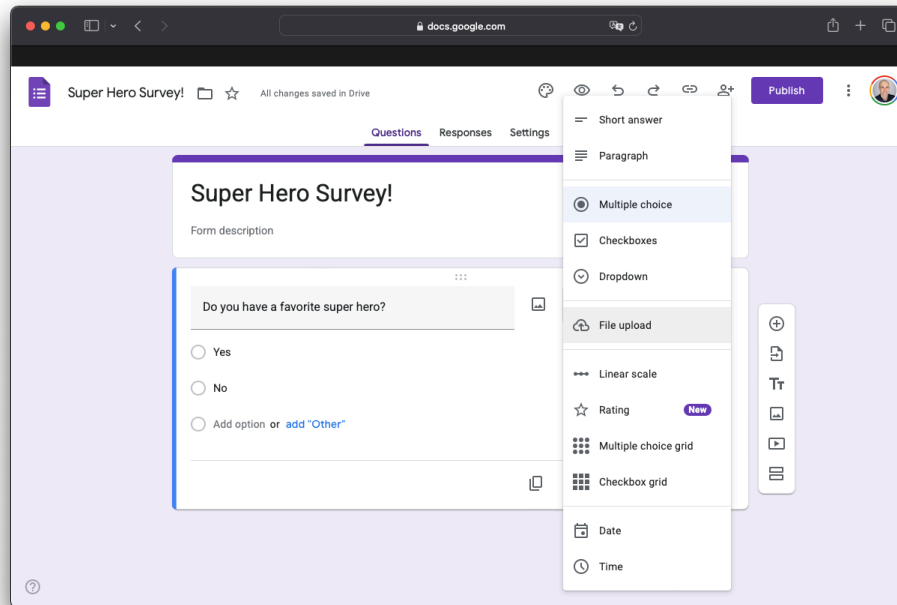
Do you have a favorite super hero?*

☐ Yes

☒ No

Next

WYSIWYG Interface



- ⊗ Reproducibility
- ⊗ Version control
- ⊗ Limited features
- ⊗ Open source

Why not make surveys
from **code**?

~~Limitations~~

Features

- ✓ Reproducibility
- ✓ Version control
- ✓ Lots of features
- ✓ Open source

Introducing **surveydown**!



Survey design using Quarto and Shiny



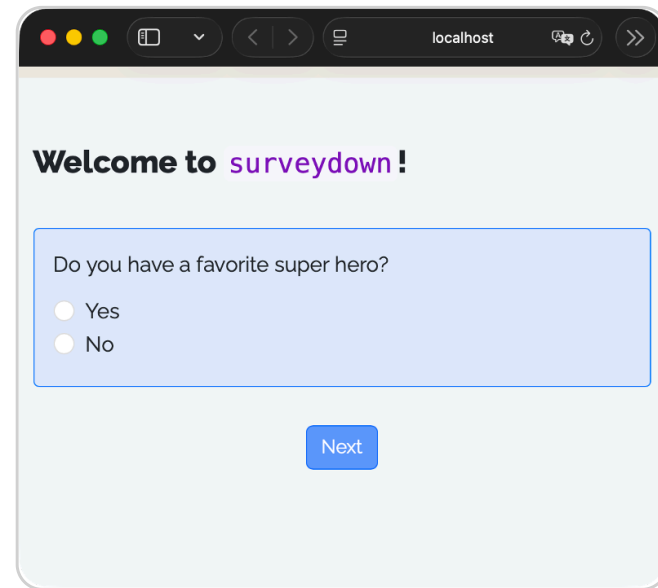
+



+



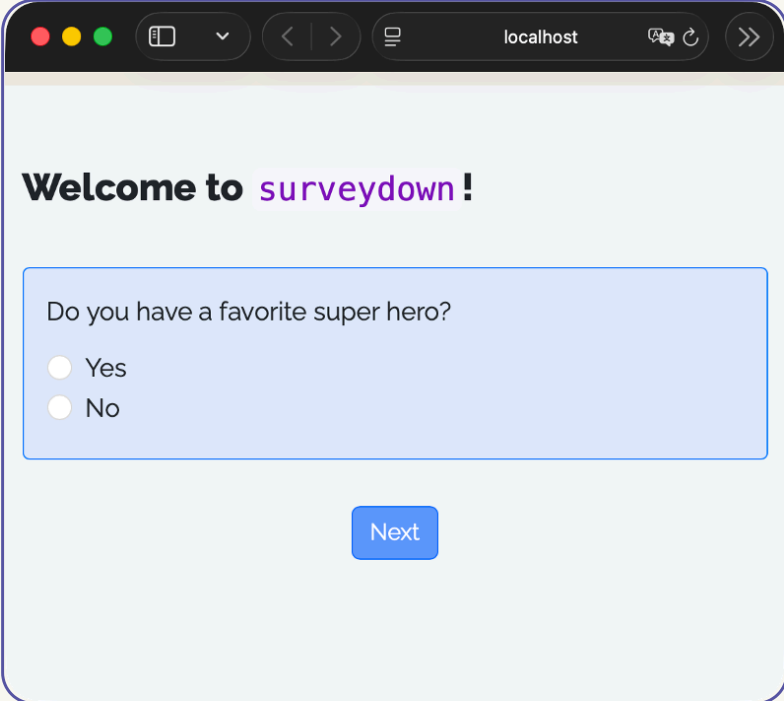
=



survey.qmd

```
1 ---
2 survey-settings:
3   use-cookies: true
4   rate-survey: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id   = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25 ```
26
27 --- some_other_page
28
29 Other content...
```

Rendered Survey



localhost

Welcome to surveydown!

Do you have a favorite super hero?

☐ Yes

☐ No

Next

But...



Code-Only Interface

No GUI. Not everyone is comfortable working purely in code.



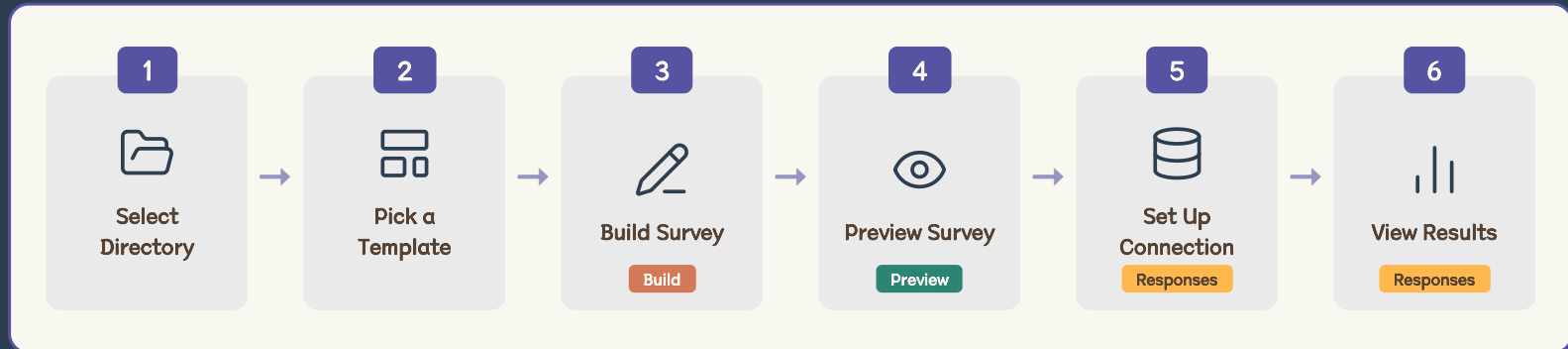
Learning Curve

The survey pipeline is not obvious. It takes time to learn.

Introducing **sdstudio**!



Studio Workflow



Launch with a simple command

```
sdstudio :: launch()
```

surveydown Studio

Build

Preview

Responses

127.0.0.1

+

Create New Survey

Template:

Default

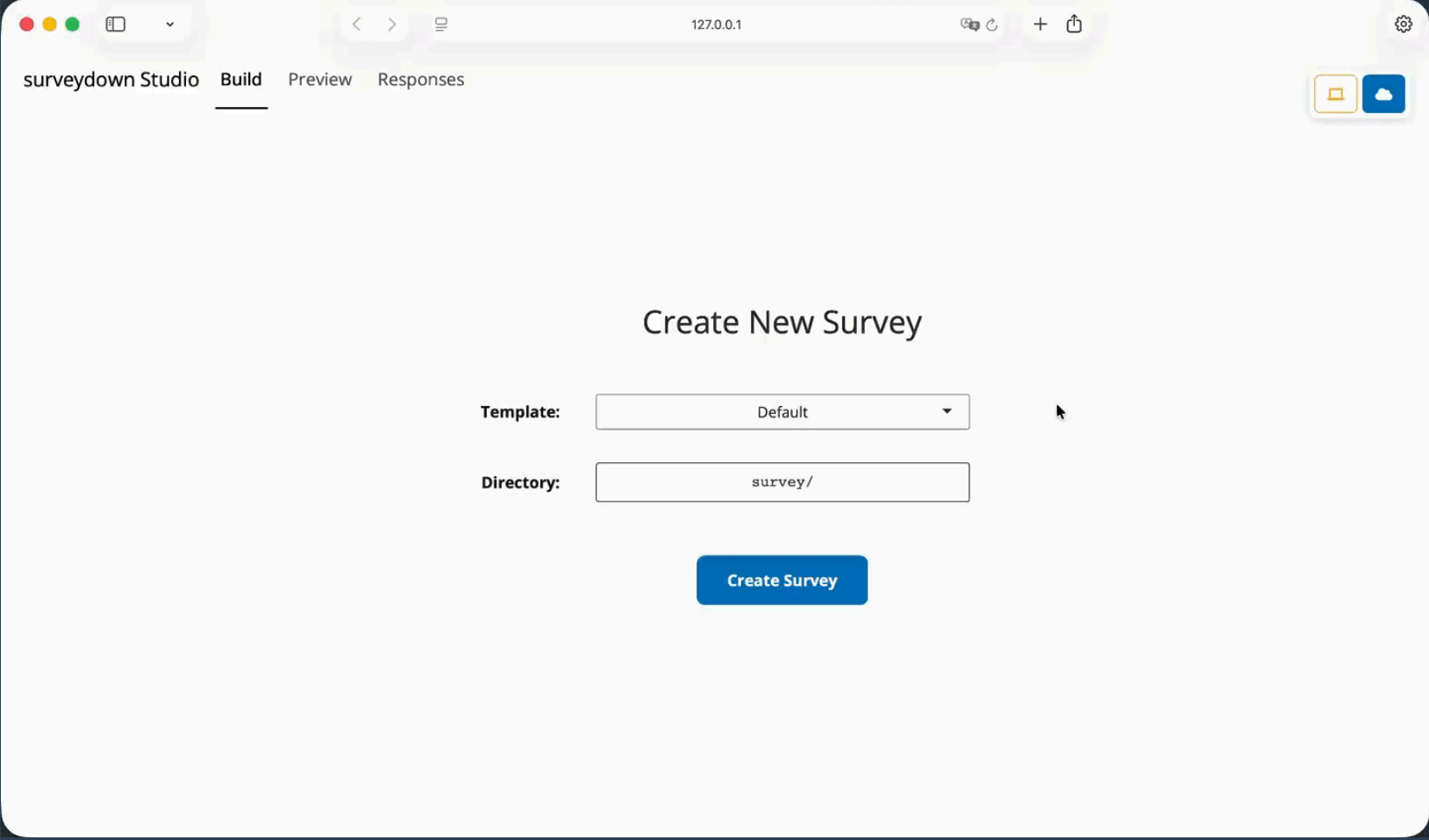
Directory:

survey/

Create Survey



Start with a Template



The screenshot shows a web browser window with the address bar displaying '127.0.0.1'. The page title is 'surveydown Studio'. The navigation bar includes 'Build' (active), 'Preview', and 'Responses'. In the top right corner, there are icons for a laptop, a cloud, and a settings gear. The main content area is titled 'Create New Survey' and contains two form fields: 'Template:' with a dropdown menu set to 'Default', and 'Directory:' with a text input field containing 'survey/'. Below these fields is a blue button labeled 'Create Survey'.

surveydown Studio Build Preview Responses

Create New Survey

Template: Default

Directory: survey/

Create Survey

surveydown.org/templates

surveydown.org

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Basic

Default Template

Question Types

Questions Defined with YAML

Conditions

Conditional Showing

Conditional Skipping

Conditional Stopping

Randomization

Random Options

Random Options Predefined

Option Shuffling

Reactivity

Reactive Questions

Reactive drilldown

External Redirect

Live Polling

Conjoint

Conjoint Buttons

Conjoint Tables

Custom

Custom Leaflet Map

Custom Plotly Chart

Templates

To help you get started with surveydown, we created some templates that cover many of the features of the package. Feel free to use any as a starting point for your own surveys or as a reference for specific features.

Call `sd_create_survey()` with proper arguments to create your desired templates in your preferred directory. Refer to the [Start with a template](#) section for more details.

Default Template

Welcome to surveydown! This is a default template of a surveydown survey. It has two pages with one question on each page.
Here is a basic "multiple choice" question:

Which type of penguin do you like the best?
- Adelin
- Chinstrap
- Gentoo

You need to insert next buttons with `sd_next()` on each page. By default, it will go to the next page defined in the survey, but you can also specify the page with the `next_page` argument and to the name of the page you want to go to next.

Source Code

Default Template

BASIC

A minimum template for starting from scratch.

Template - Question Types

Hi! This is a showcase of different question types and question formatting that surveydown supports. Please make your choice below:

Where do you want to go to?
- Take me to the page on Question Types the next page!
- Take me to the page on Question Formatting.
- Skip to the end.

Source Code

Question Types

BASIC

A template showcasing all built-in [question types](#).

Template - Questions Defined with YAML

This is a showcase of defining questions with a `questions.yml` file.
The `sd_create_survey()` shows a `yaml` argument default to `"questions.yml"`. If you have such file in your project root folder, you can omit it to respect your favorite definition.
You can also change to other file name or directory and define questions like this:

```
sd_create_survey(
  yaml = "my_questions.yml",
  path = "folder_questions/1234_my.yml"
)
```


Please click on the "Next" button to see the demo questions.

Source Code

Questions Defined with YAML

BASIC

A template showcasing all built-in [question types](#) using a YAML file.

Categories

All (17)

Basic (3)

Conditions (3)

Conjoint (2)

Custom (2)

Randomization (3)

Reactivity (4)

Site made with   and [quarto](#)

Choose Your Directory

The screenshot shows a web application interface for 'surveydown Studio'. At the top, there is a navigation bar with tabs for 'Build', 'Preview', and 'Responses', with 'Build' being the active tab. The main content area is titled 'Create New Survey'. It features two input fields: 'Template:' with a dropdown menu set to 'Default', and 'Directory:' with a text input field containing 'survey/'. Below these fields is a prominent blue button labeled 'Create Survey'. The interface is clean and modern, with a white background and blue accents.

surveydown Studio Build Preview Responses

Create New Survey

Template: Default

Directory: survey/

Create Survey

The Build Tab

surveydown Studio Build Preview Responses

Structure

Page: welcome

- T # Default Template Welcome to...
- Q penguins | mc
What's your favorite penguin?
- T There are several ways to...

Page: page2

- T This is another page in...
- Q silly_word | text
Write a silly word here:
- T On this page, we don't...

Page: end

- T ## End This is the...

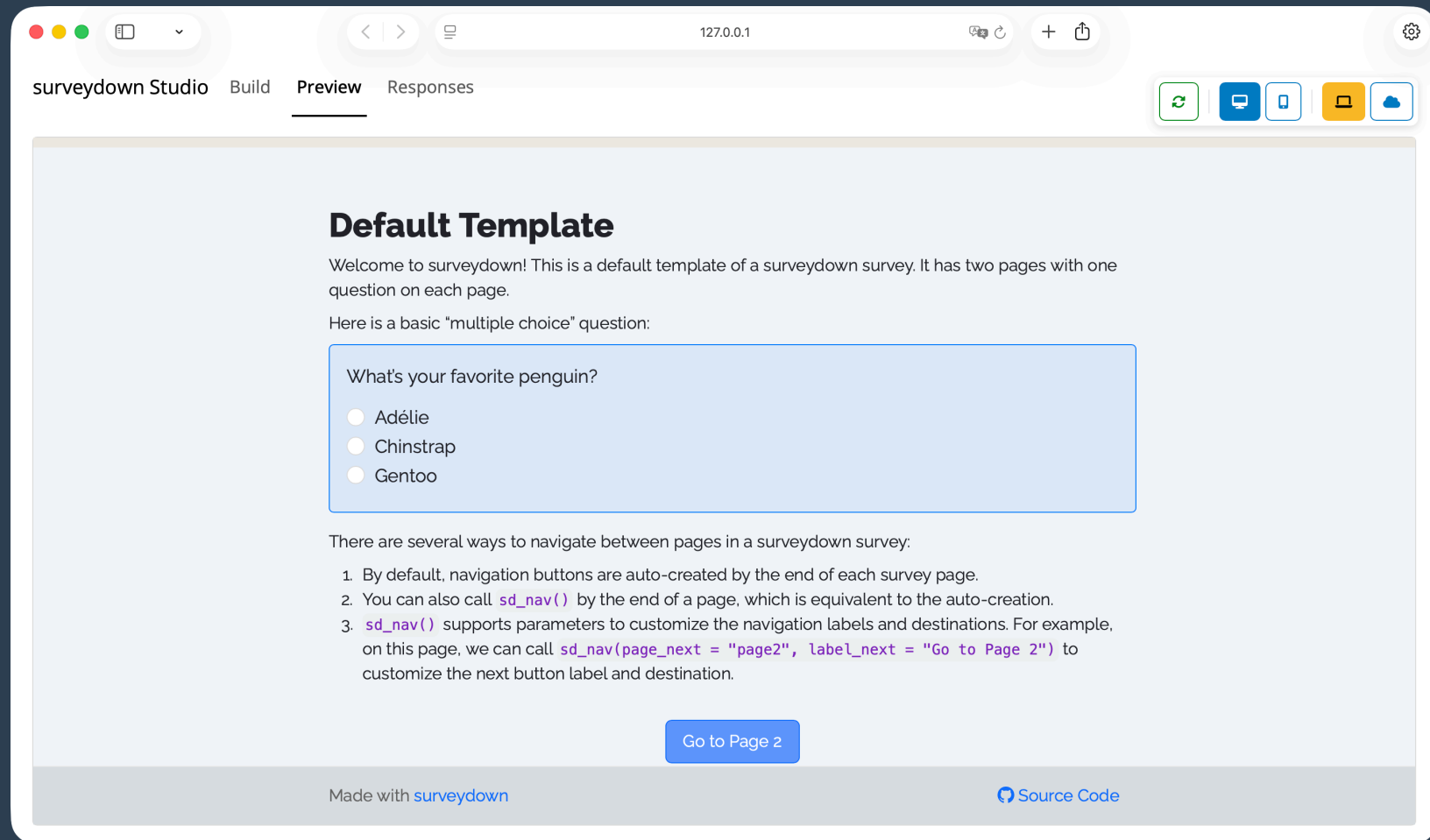
Add A New Page

Code

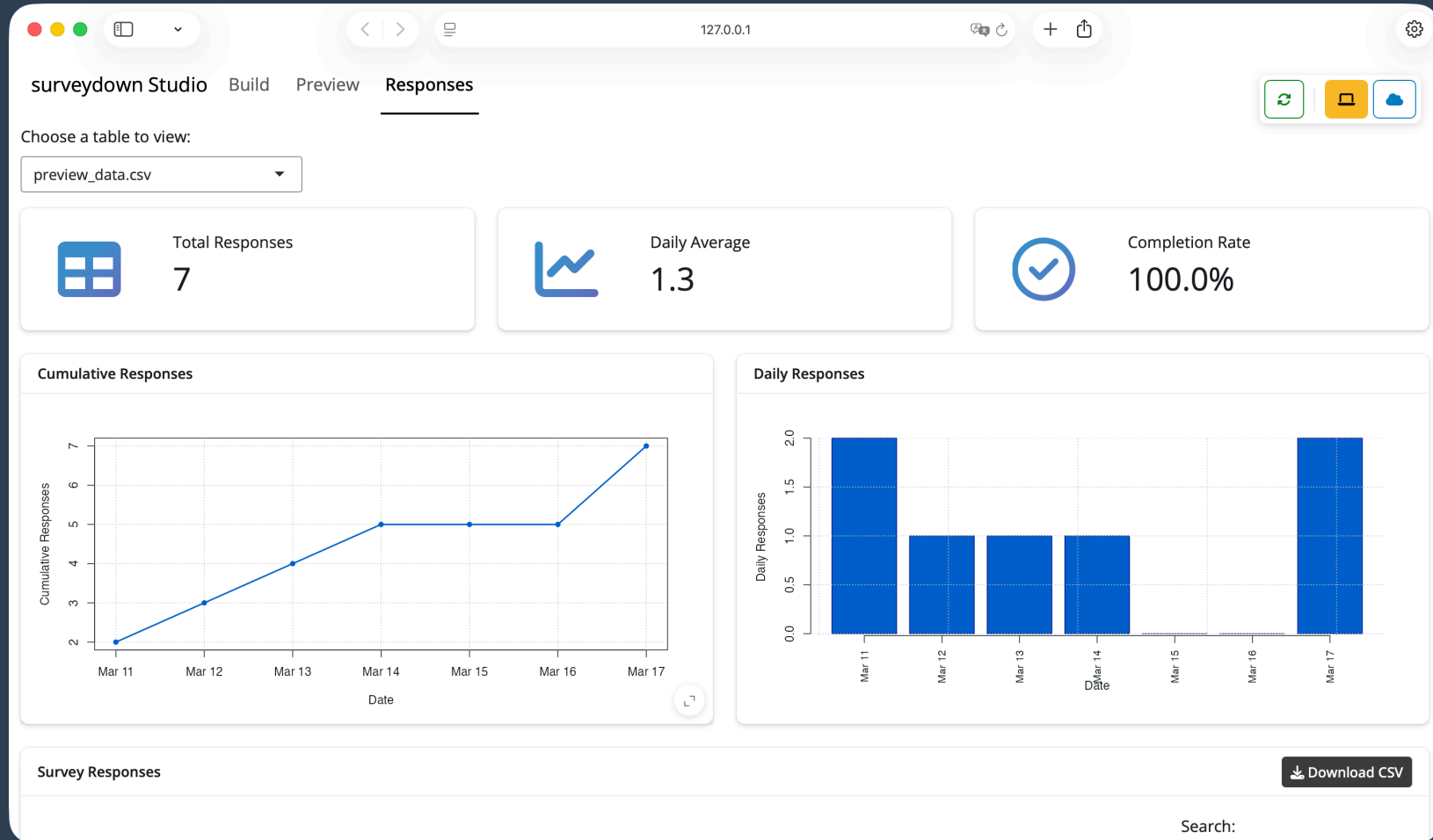
survey.qmd app.R

```
37 --- welcome
38
39 # Default Template
40
41 Welcome to surveydown! This is a default template of a surveydown survey. It has two
42 pages with one question on each page.
43 Here is a basic "multiple choice" question:
44
45 ```{r}
46 sd_question(
47   type = 'mc',
48   id   = 'penguins',
49   label = "What's your favorite penguin?",
50   option = c(
51     'Adélie' = 'adelie',
52     'Chinstrap' = 'chinstrap',
53     'Gentoo' = 'gentoo'
54   )
55 )
56 ```
57
58 There are several ways to navigate between pages in a surveydown survey:
59
60 1. By default, navigation buttons are auto-created by the end of each survey page.
61 2. You can also call `sd_nav()` by the end of a page, which is equivalent to the
62    auto-creation.
63 3. `sd_nav()` supports parameters to customize the navigation labels and destinations.
64    For example, on this page, we can call `sd_nav(page_next = "page2", label_next = "Go to
65    Page 2")` to customize the next button label and destination.
66
67 ```{r}
```

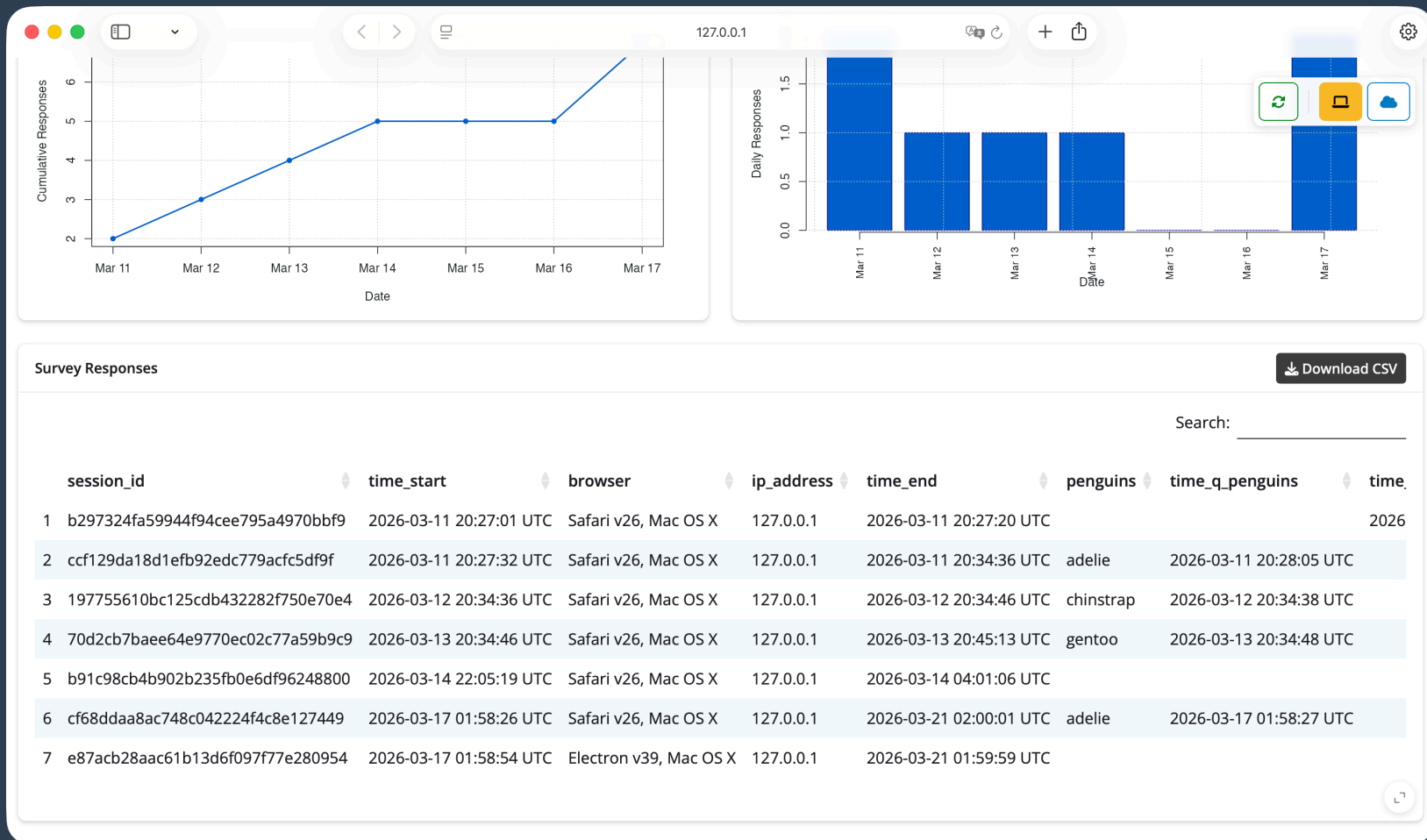
The Preview Tab



The Responses Tab

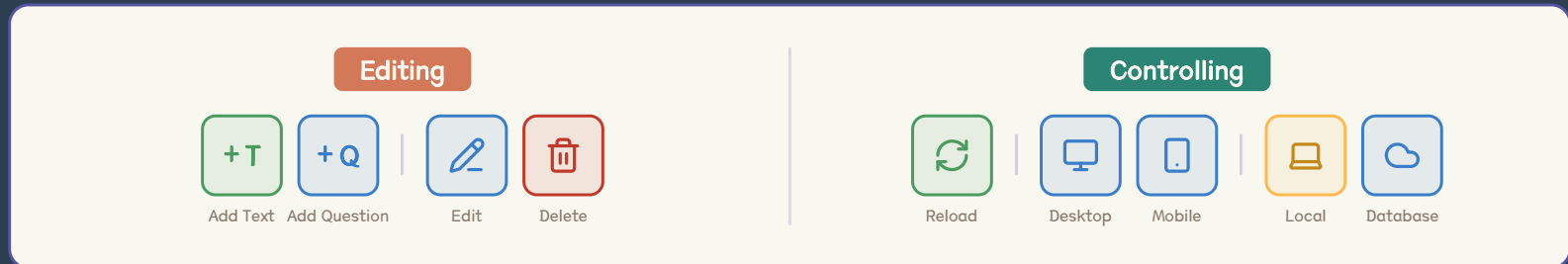


The Responses Tab - Continued

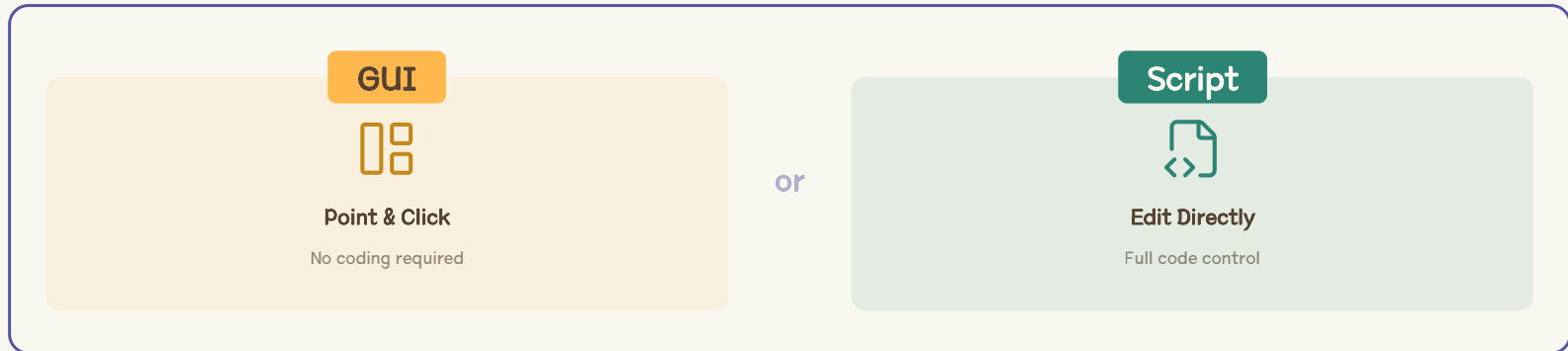


Studio Features

Self-explanatory Icons



Editing



GUI - Toggling

The screenshot displays the surveydown Studio application window. The top navigation bar includes 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Structure' panel on the left shows a hierarchical view of the survey pages. It includes 'Page: page1' with a title 'Some texts.' and a multiple-choice question 'penguins | mc' with the label 'What's your favorite penguin?'. Below this is 'Page: end' with the text '## End This is the...'. An 'Add A New Page' button is at the bottom of the Structure panel. The 'Code' panel on the right shows the R code for the survey, with tabs for 'survey.qmd' and 'app.R'. The code includes library loading, page definitions, question definitions, and closing the survey.

```
survey.qmd  app.R

33 library(surveydown)
34 ---
35
36
37 --- page1
38
39 # Title
40
41 Some texts.
42
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ---
55
56 --- end
57
58 ## End
59
60 This is the last page in the survey.
61
62 ```{r}
63 sd_close()
64 ---
```

GUI - Content Editing

The screenshot displays the surveydown Studio interface, which is used for creating and editing surveys. The interface is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1**
 - # Title** Some texts.
 - penguins | mc**
What's your favorite penguin?
- Page: end**
 - ## End** This is the...

At the bottom of the Structure panel, there is a button labeled "Add A New Page".

Code Panel:

The Code panel shows the R code for the survey, with the file name "survey.qmd" and the script name "app.R". The code is as follows:

```
33 library(surveydown)
34 
35 
36 
37 --- page1
38 
39 # Title
40 
41 Some texts.
42 
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 
55 
56 --- end
57 
58 ## End
59 
60 This is the last page in the survey.
61 
62 ```{r}
63 sd_close()
64 
```

GUI - Content Addition

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- At the top, there are tabs for "surveydown Studio", "Build", "Preview", and "Responses".
- The "Structure" panel shows a hierarchical view of the survey content.
 - Page: page1** (selected):
 - A text block: "# Title New texts."
 - A multiple-choice question: "penguins | mc" with the label "What's your favorite penguin?".
 - Page: end**:
 - A text block: "## End This is the..."
- At the bottom of the Structure panel is a button labeled "Add A New Page".

Code Panel:

- The "Code" panel shows the R code for the survey, with tabs for "survey.qmd" and "app.R".
- The code in "survey.qmd" is as follows:

```
33 library(surveydown)
34 ---
35
36
37 --- page1
38
39 # Title
40
41 New texts.
42
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ---
55
56 --- end
57
58 ## End
59
60 This is the last page in the survey.
61
62 ```{r}
63 sd_close()
64 ---
```

GUI - Page Addition

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel: This panel shows the hierarchical structure of the survey. It includes a top bar with navigation tabs: "surveydown Studio", "Build", "Preview", and "Responses". Below this, the "Structure" tab is active, showing a list of pages and questions. The first page is "Page: page1", which contains two questions: "penguins | mc" (a multiple-choice question) and "text_question | text" (a text question). The second page is "Page: end", which contains a single question: "## End This is the...". At the bottom of the Structure panel, there is a button labeled "Add A New Page".

Code Panel: This panel shows the R code for the survey. It includes a top bar with tabs for "survey.qmd" and "app.R". The "survey.qmd" tab is active, showing the following R code:

```
41 new texts.  
42  
43 ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 ```  
55 ```{r}  
56 sd_question(  
57   type = "text",  
58   id = "text_question",  
59   label = "Write something:"  
60 )  
61 ```  
62  
63  
64 --- end  
65 ## End  
66  
67  
68 This is the last page in the survey.  
69  
70 ```{r}  
71 sd_close()  
72 ```
```

GUI - Page Relocation

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel: This panel shows the hierarchical structure of the survey. It includes three pages: **Page: page1**, **Page: page2**, and **Page: end**. Each page contains a list of elements with their respective icons and labels. For example, **Page: page1** contains a text element labeled **# Title New texts.**, a multiple-choice question labeled **penguins | mc** with the text **What's your favorite penguin?**, and a text question labeled **text_question | text** with the text **Write something:**. The **Page: end** section contains a text element labeled **## End This is the...**. At the bottom of the Structure panel, there is a button labeled **Add A New Page**.

Code Panel: This panel shows the R code for the survey. The code is organized into sections corresponding to the pages in the Structure panel. The **survey.qmd** file contains the following code:

```
41 new texts.
42
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ```
55 ```{r}
56 sd_question(
57   type = "text",
58   id = "text_question",
59   label = "Write something:"
60 )
61 ```
62
63 --- page2
64
65 This is page 2.
66
67
68 --- end
69
70 ## End
71
72
```

GUI - Content Relocation

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1**
 - Text: # Title New texts.
 - MC: penguins | mc
What's your favorite penguin?
 - Text: text_question | text
Write something:
- Page: page2**
 - Text: This is page 2.
- Page: end**
 - Text: ## End This is the...

At the bottom of the Structure panel is a button labeled "Add A New Page".

Code Panel:

The Code panel shows the R code for the survey, with tabs for "survey.qmd" and "app.R". The visible code is as follows:

```
41 new texts.  
42  
43 ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 ```  
55 ```{r}  
56 sd_question(  
57   type = "text",  
58   id   = "text_question",  
59   label = "Write something:"  
60 )  
61 ```  
62  
63 --- page2  
64 This is page 2.  
65  
66 --- end  
67  
68 ## End  
69  
70  
71  
72
```

Script Editing

The screenshot displays the surveydown Studio web application. The interface is divided into two main panels: 'Structure' on the left and 'Code' on the right. The top navigation bar includes 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Structure' panel shows a hierarchical view of the survey pages. It includes three pages: 'Page: page1', 'Page: page2', and 'Page: end'. Each page contains a list of elements with their respective icons and edit/delete buttons. 'Page: page1' has a title element and a multiple-choice question about penguins. 'Page: page2' has a text element and a text question. 'Page: end' has a closing text element. An 'Add A New Page' button is at the bottom. The 'Code' panel shows the R script for the survey, with tabs for 'survey.qmd' and 'app.R'. The script defines the survey structure, including the multiple-choice question and the text question, and ends with a closing command.

surveydown Studio Build Preview Responses

Structure

Page: page1

- # Title New texts.
- penguins | mc
What's your favorite penguin?

Page: page2

- This is page 2.
- text_question | text
Write something:

Page: end

- ## End This is the...

Add A New Page

Code

```
survey.qmd app.R
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ---
55 --- page2
56
57 This is page 2.
58
59 ```{r}
60 sd_question(
61   type = "text",
62   id = "text_question",
63   label = "Write something:"
64 )
65 ---
66
67 --- end
68
69 ## End
70
71 This is the last page in the survey.
72
73 ```{r}
74 sd_close()
75
```

Instant Preview

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

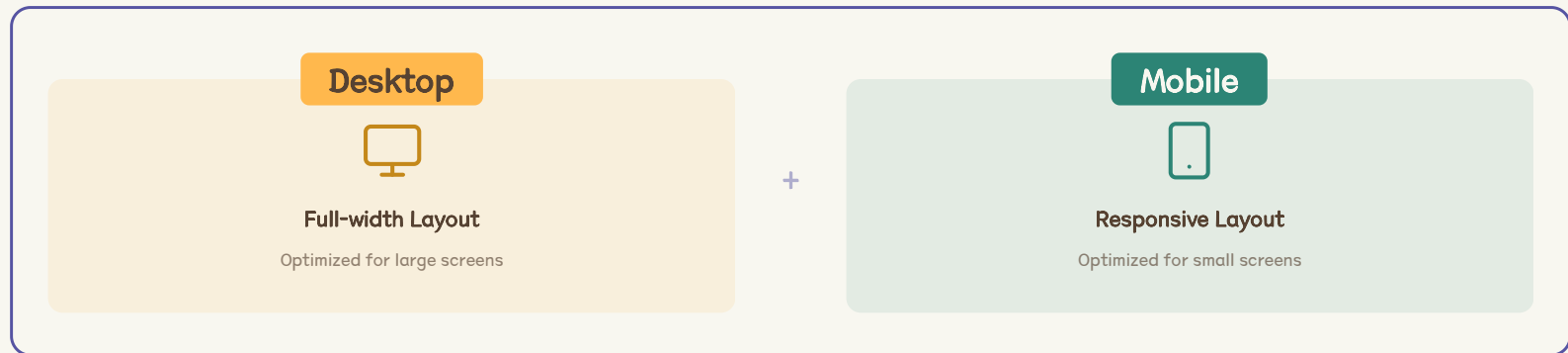
- Page: page1**
 - Text question: # Title New texts.
 - Multiple choice question: penguins | mc. Label: What's your favorite penguin?
- Page: page2**
 - Text question: This is page 2.
 - Text question: text_question | text. Label: Write something:
- Page: end**
 - Text question: ## End This is the...

Code Panel:

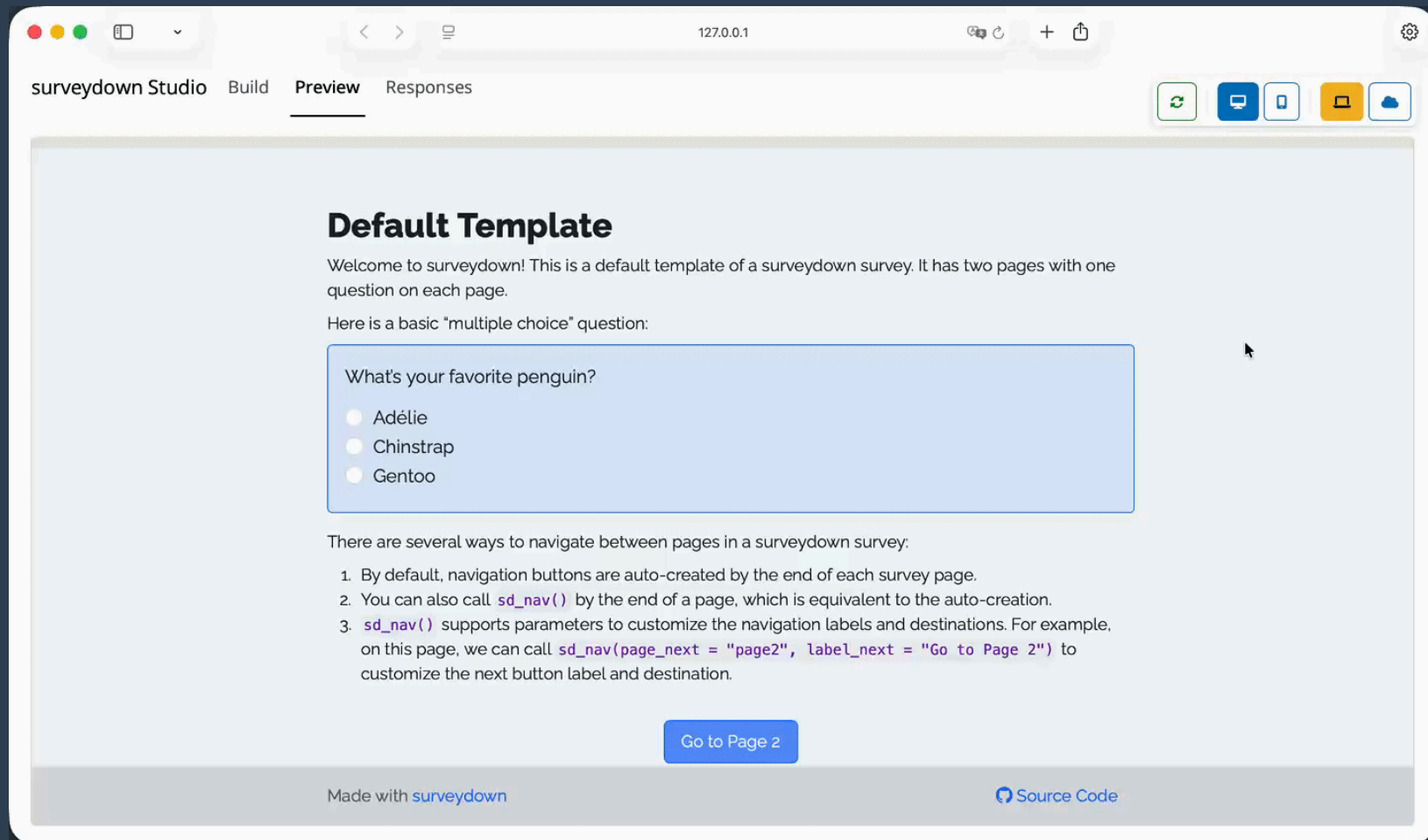
The Code panel shows the R code for the survey, with tabs for survey.qmd and app.R. The visible code is as follows:

```
34 ----
35
36
37 --- page1
38 # Title
39
40
41 New texts.
42
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ----
55 --- page2
56
57 This is page 2.
58
59 ```{r}
60 sd_question(
61   type = "text",
62   id = "text_question",
63   label = "Write something:"
64 )
65 ----
```

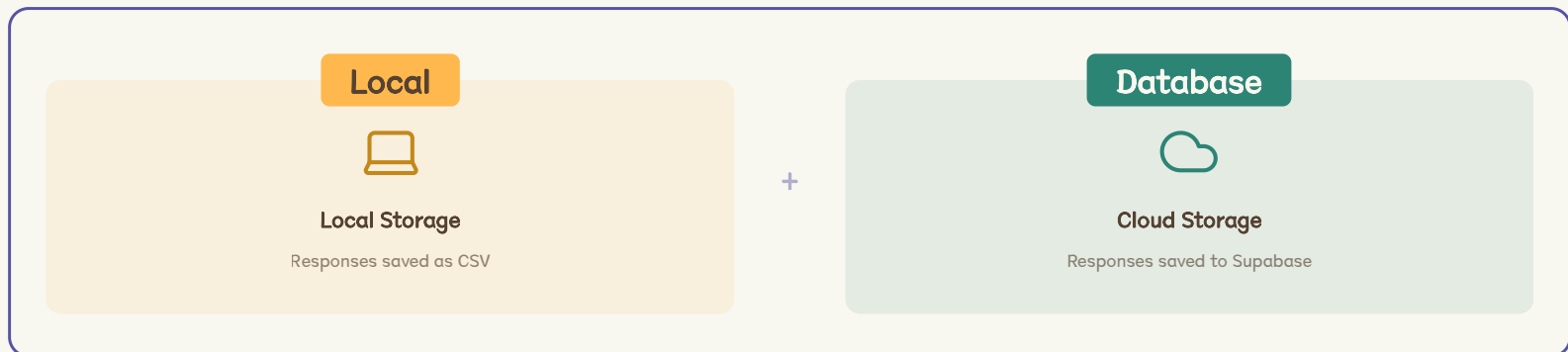
Views



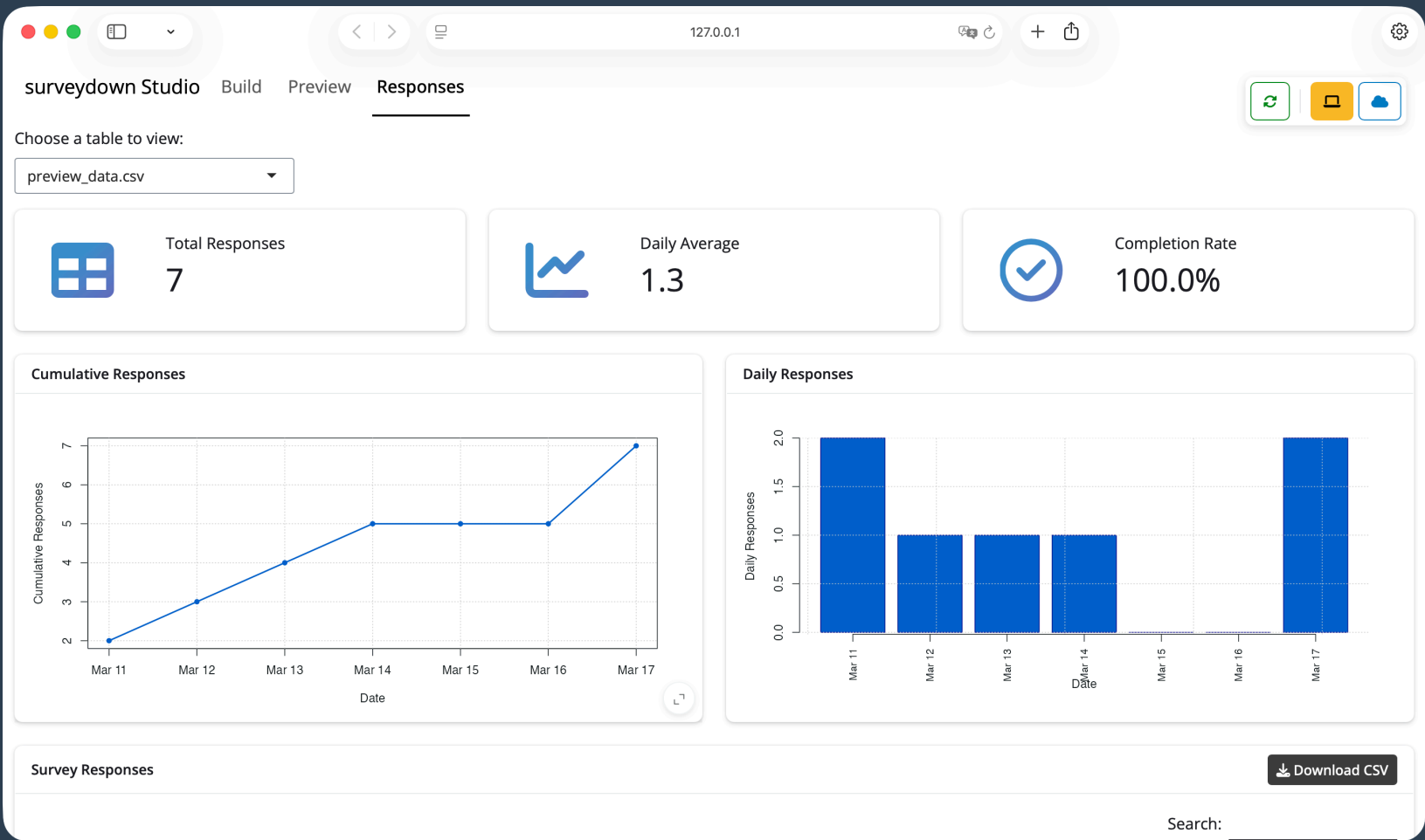
Desktop and Mobile Views



Responses



Local Responses



Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Settings

Not connected

Choose a table to view:

Select or type table name...

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Settings

Not connected

Choose a table to view:

Select or type table name...

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Close

Not connected

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

default

Test Connection

Choose a table to view:

Select or type table name...

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Close

Connected (GSSAPI disabled)

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

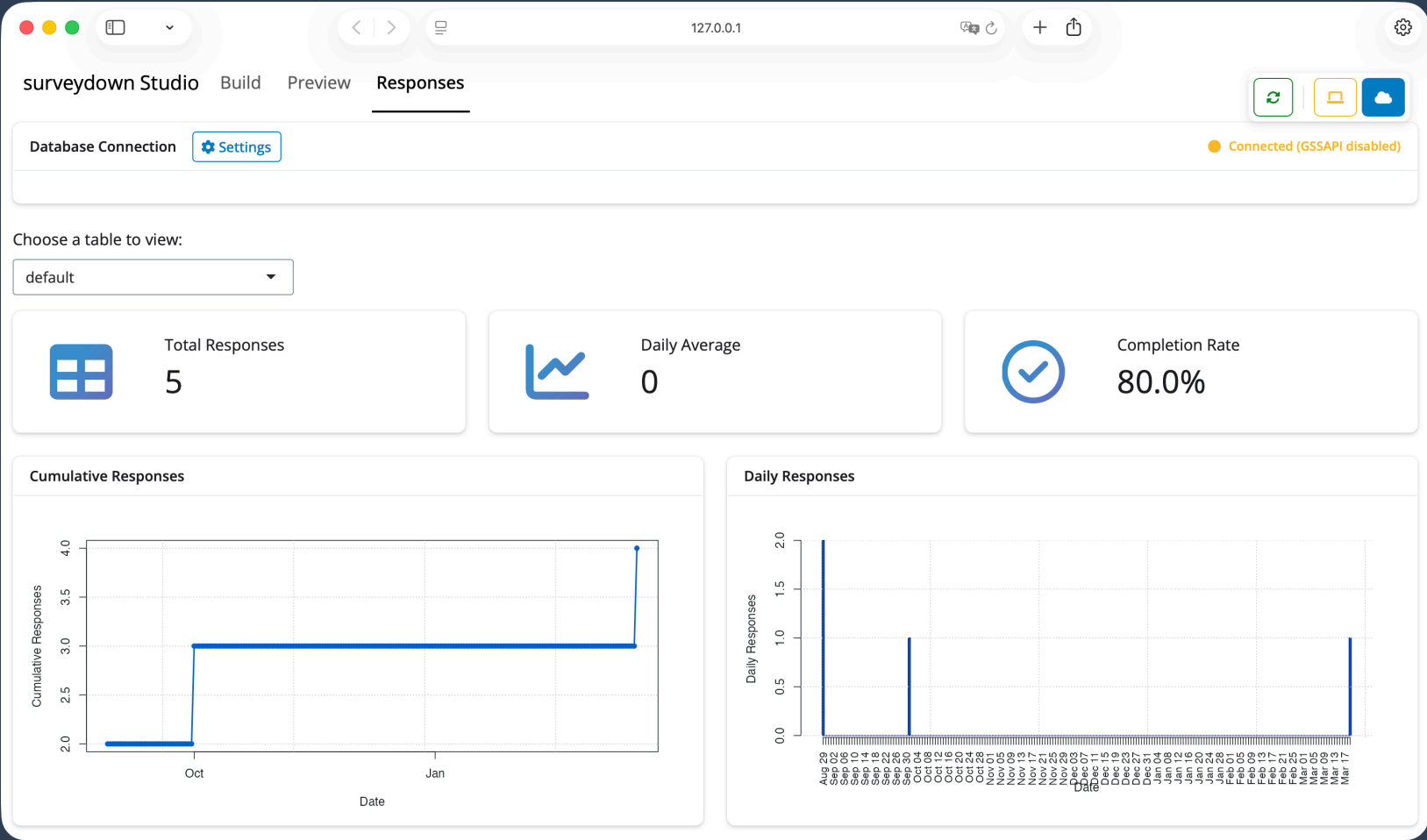
default

Test Connection

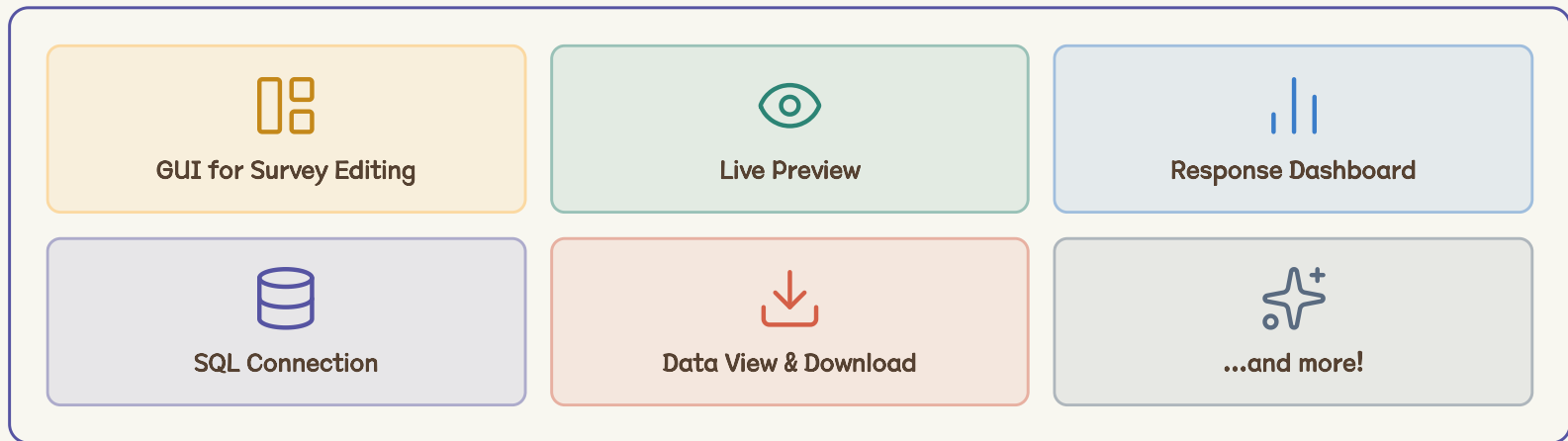
Choose a table to view:

default

Database Responses



sdstudio gives you...



surveydown + sdstudio

surveydown

 Text-based editing

 Published paper

 8000+ CRAN downloads

sdstudio

 GUI editing & live preview

 Actively developed

Open to contributions

 Better together!

Ways to help:

github.com/surveydown-dev/sdstudio/

- Try it out!
- Give it a ★
- Post an issue
- Join the GitHub discussion
- Contribute new features



Meet the team!



John Helveston, Ph.D.



Pingfan Hu



Bogdan Bunea

surveydown.org

